Simignore: Enhancing Multimodal Large Language Models Complex Reasoning via Similarity Computation

Anonymous EMNLP submission

Abstract

Multimodal large language models have experienced rapid growth, and numerous different models have emerged. The interpretability of MLLM remains an under-explored area. Especially when faced with more complex tasks such as chain-of-thought reasoning, its internal mechanisms still resemble a black box that is difficult to decipher. By studying the interaction and information flow between images and text, we noticed that in models such as LLaVA1.5, image tokens that are semantically related to text are more likely to have information flow convergence in the LLM decoding layer, and these image tokens receive higher attention scores. However, those image tokens that are less relevant to the text do not have information flow convergence, and they only get very small attention scores. To efficiently utilize the image information, we propose a new image token reduction method, Simignore, which aims to improve the complex reasoning ability of MLLM by computing the similarity between image and text embeddings and ignoring image tokens that are irrelevant and unimportant to the text. Through extensive experiments, we demonstrate the effectiveness of our method for complex reasoning tasks.

1 Introduction

Multimodal Large Language Modeling (MLLM) has made rapid development in recent years and become a research hotspot in the field of computer vision as well as natural language processing. Multimodal large language models (LLMs) have shown impressive performance on complex reasoning by leveraging chain-of-thought (CoT) prompting to generate intermediate reasoning chains as the rationale to infer the answer. MLLM utilizes a visual encoder such as CLIP to process picture patches to get visual tokens, which are used as the context of visual information to accomplish visual-textual reasoning tasks. The visual coder processes these

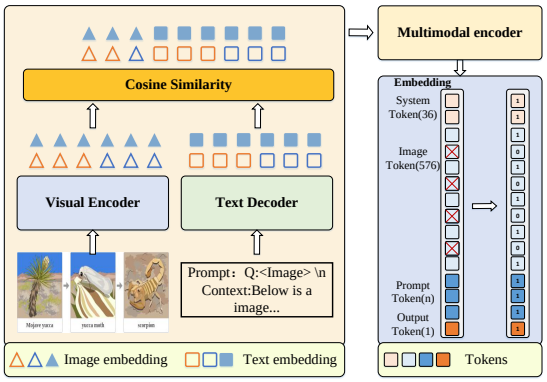


Figure 1: The simplified structure of our method.

image patches into hundreds of tokens, e.g., 576 for CLIP (Radford et al., 2021) and 729 for siglip (Zhai et al., 2023). These extra image tokens lead to an increase in computation, and some studies (Chu et al., 2023a; Yuan et al., 2023) have reduced the inference cost by using smaller LLMs with fewer parameters, however, these approaches also lead to a decrease in the inference power of LLMs (Chu et al., 2024). Therefore, a better approach is to reduce the computational cost by decreasing the length of input tokens, and at the same time, studies (Shang et al., 2024; Chen et al., 2024) have shown that not all image tokens are important for model inference and that a proper reduction of image tokens can improve the inference ability of LLMs, leading to a significant increase in the accuracy of LLMs on visual complex reasoning tasks.

There are some works on reducing image tokens, FastV (Chen et al., 2024) visualized the attention scores of system tokens, image tokens, and user tokens during the LLM inference process and found that the weight of image tokens was very small after the second layer, so he chose to discard half of the image tokens with lower scores at the second layer based on the ordering of their attention scores. LLaVA-PruMerge (Shang et al., 2024) utilized the self-attention mechanism between category tokens

and visual tokens to observe the distribution of attention between them and found that most of the visual tokens had attention values close to zero with the category tokens, indicating that these tokens are not critical in the image representation. Therefore he selects appropriate image tokens based on the spatial similarity between visual tokens and CLS tokens, then clusters the pruned tokens and maintains the completeness and richness of the visual information by merging the clustered tokens with the unpruned ones.

Regarding the above two methods of reducing image tokens, although the two methods of image markup approximation have achieved good results, there are limitations to their methods: (1) the above two methods of reducing image tokens do not take into account the interaction with textual prompt, and it is quite easy to filter out the tokens related to the "semantics" of the text. (2) At present, the interaction between image tokens and text tokens is still unclear, and how LLM utilizes image tokens for answering is still unknown and needs further exploration.

We employ information flow to explore the interaction between image tokens and text prompts. We define information flow as the process by which image tokens gradually converge on the semantics associated with text during processing in the attention mechanism of image tokens. Specifically, we visualize the attention score of the image token in the LLM decoder and superimpose it on the image patch. We find a convergence of information flow on the image patch related to the text. As shown in Fig. 4, we can observe that the image patches related to the text options, such as mushroom and bilberry, converge information flow in the network, and they get higher attention scores. In other cases, when answering a question does not require a reference image, the network does not pay much attention to the image and the information flow does not converge.

Therefore, we design the image-text token filtering algorithm named Simignore to keep irrelevant tokens from causing interference. Specifically, as shown in Fig. 1, we map the embeddings of image tokens and prompt tokens to the same similarity metric space, in which all tokens are points on a two-dimensional plane. Then, we compute the similarity value between the image token and text token using the similarity algorithm, we select the K image tokens with the highest similarity and record their subscripts. Finally, we keep the selected image tokens, and for the unselected tokens, we set their corresponding attention mask to 0 to ignore them.

In summary, the contribution of this work is as follows:

- We found through information flow that in the LLM decoder, images that are semantically related to textual markups are more likely to have a convergence of information flow.
- We propose a new method, Simignore, which aims to retain the image tokens that interact with the text and ignore the irrelevant and unimportant image tokens to improve the ability of MLLM in complex reasoning tasks.
- Through extensive experiments, we demonstrate the effectiveness of our method for complex reasoning in visual reasoning tasks with different MLLMs.

2 Related work

2.1 Multimodal large language models

Multimodal large language models can process and understand data from many different modalities, such as text, and images, and thus demonstrate excellent performance in a variety of tasks.CLIP (Radford et al., 2021) and BLIP (Li et al., 2022) use a pre-training approach that maps images and text into the same feature space, enabling the model to correlate images and text by comparing feature vectors.LLaVA (Liu et al., 2023), MiniGPT-4 (Zhu et al., 2023), Qwen-VL (Bai et al., 2023), and CogVLM (Wang et al., 2023) use pre-trained ViT to process information from images.

2.2 Implicit reasoning and chains of thought

Implicit reasoning in LLM refers to the ability of a model to generate responses and perform tasks without providing explicit intermediate steps. Researchers have enhanced this capability by exploring different architectures and training methods. For example, GPT-2 (Radford et al., 2019) and GPT-3 (Brown et al., 2020) implicitly understand the context and generate coherent text by training a corpus and more sophisticated reasoning capabilities, respectively. The concept of thought chaining, on the other hand, mimics human problem-solving by generating a series of logical steps that lead to

the final output.Wei (Wei et al., 2022) proposed a "thought chaining" cueing approach to significantly improve the performance of LLMs on complex reasoning tasks by generating intermediate inference steps. Kojima et al. Kojima (Kojima et al., 2022) further showed that simple prompts allow LLMs to improve their reasoning ability by thinking step-by-step.

2.3 Image-Text Multimodal Similarity

Image-text similarity research is a key area in multimodal learning for assessing the consistency between images and text, and the accuracy of cross-modal retrieval has been significantly improved in recent years by a variety of methods. These methods include the joint embedding technique (Frome et al., 2013; Kiros et al., 2014; Faghri et al., 2017), the SGRAF network proposed by Diao (Diao et al., 2021). the instance comparison embedding method by Zeng (Zeng et al., 2024). the unified comparison learning method by Yang (Yang et al., 2022). and the image feature and class semantic embedding in cosine metric space by Liu (Liu et al., 2021). In particular, cosine similarity is used to recognize image tokens with strong relevance to textual information due to its advantages in reducing intra-class variance and improving recognition ability.

3 The Proposed Method

To present a visualization of the information flow, we employ the Attention Score techniques to gain a comprehensive understanding of the information flow. The Attention Score reveals the forward processing of the model, showing the contribution of different input elements to the final output.

3.1 Influence rate of image token on output token

For the output token of the complex reasoning task such as the ScienceQA dataset (Lu et al., 2022), in the n-th layer, we define $\mathcal{G}$ as the indices set of all tokens and $\mathcal{G}$ can be divided into three parts that represent the indices set of system, image, and user tokens:

$$\mathcal{G} = \mathcal{S} + \mathcal{I} + \mathcal{U}, \tag{1}$$

where $\mathcal{S} = \{1, \ldots, N_{\text{sys}}\}$ represents the index of system token, N_{sys} represents the length of system token, $\mathcal{I} = \{N_{\text{sys}} + 1, \ldots, N_{\text{sys}} + N_{\text{img}}\}$ represents the index of image token, N_{img} represents the length of image token, and $\mathcal{U} = \{N_{\text{sys}} + N_{\text{img}} + 1, \ldots, N_{\text{sys}} + N_{\text{img}} + N_{\text{user}}\}$ represents the index of user token, N_{user} represents the length of user token. $A_{i,j}$ is defined as the total attention score of the output token's attention on different types of tokens. For the i-th query token, the attentions from system, image, and user tokens are summed as 1:

$$\sum_{j \in \mathcal{S}} A_{i,j} + \sum_{j \in \mathcal{I}} A_{i,j} + \sum_{j \in \mathcal{U}} A_{i,j} = 1, \tag{2}$$

To ensure that the sum of attention scores for each token is 1, it is necessary to normalize the above summation results to calculate the total attention score for the image token:

$$\lambda_{\text{img}}^{j} = \sum_{j \in \mathcal{I}} A_{i,j}. \tag{3}$$

There are 576 image tokens in LLaVA1.5. The shape of the attention mask matrix is (B, H, N_{img}, N_{img}), we first perform unsequenced to change the attention matrix to (H, N_{img}, N_{img}), the attention matrix is shown in the Fig. 4 and Fig. 3, the horizontal coordinate stands for Q, the vertical coordinate stands for K, and the first row of the vertical coordinate stands for System token, image token, and the influence rate of user token on output token, taking the image token i.e. id from 35-611, we can get the value of 1×576 dimensions, which is the attention score corresponding to 576 tokens, and then we can change the attentions- score reshaped into a 14×14 patch superimposed on the original graph, i.e., the heat map of the influence rate.

Through the influence-score heat map, we can find that the image will have obvious convergence on the options appearing in the prompt, so a way to remove redundancy and keep effective image information is to do the similarity between the image and text token.

3.2 Compute similarity between image and text embeddings

In this section, we will detail our approach to focus on image information related to text. First of all, it is necessary to introduce image embedding and text embedding, and the specific flow is shown in the left part of Fig. 2.

In the inference process, the input image is $X \in \mathbb{R}^{C \times H \times W}$, where C denotes the number of channels and $H \times W$ denotes the size of the image. It is then input to the Visual and Language

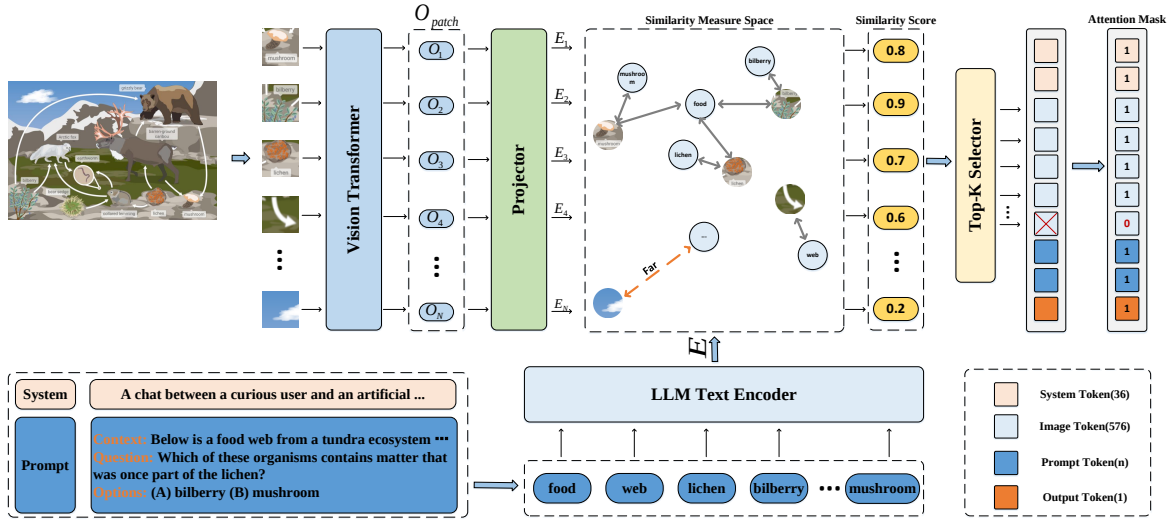


Figure 2: The holistic framework for an approach to enhance complex reasoning in multimodal large language models through similarity computation between image and text embeddings. We map the embeddings of image token and prompt token to the same similarity metric space, a process that involves operations such as regularization. Here we compute their similarity values. Then we select the K image tokens with the highest similarity and consider them important. For unselected tokens, we ignore them by setting their attention mask to 0.

Pre-training model (VLP) for processing:

$$\mathbb{F} : \mathbb{R}^{C\times H\times W} \rightarrow \mathbb{R}^{C'\times H'\times W'} \tag{4}$$

where F is the convolution operation, C' is the new number of channels, H' and W' are the new spatial dimensions. Next, the convolved feature mapping is downscaled to a specific size $N_{img} \times I$ using adaptive pooling, function adaptive-pooling $\mathbb{AP}$ is defined:

$$\mathbb{AP} : \mathbb{R}^{C'\times H'\times W'} \rightarrow \mathbb{R}^{N_{img}\times I} \tag{5}$$

The whole process is:

$$X' = \mathbb{AP}(\mathbb{F}(X)) \tag{6}$$

where $X' \in \mathbb{R}^{N_{img}\times I}$ represents the features of the image.

The dimension of the text after embedding is $TextEmb \in \mathbb{R}^{N_{usr}\times T}$. We removed the system tokens from the text and kept only the context, question, and option tokens. N_{usr} denotes the length of the text token and T denotes the dimension of the text embedding. To fuse image information with text information, it is also necessary to embed image and text into the same feature space. So We usually need to align the dimensions of the image features $N_{img} \times I$ with the dimensions of the text features $N_{img} \times T$, function feature-alignment $\mathbb{FC}$ is defined:

$$\mathbb{FC} : \mathbb{R}^{N_{img}\times I} \rightarrow \mathbb{R}^{N_{img}\times T} \tag{7}$$

$$ImgEmb = \mathbb{FC}(X') \tag{8}$$

where $ImgEmb \in \mathbb{R}^{N_{img}\times T}$ with the same feature dimensions as text. At this point, we already have image token and text token embeddings, and next, we describe our approach in detail, as shown in the right part of Fig. 2. To compute the correlation between image embedding and text embedding, we first normalize $ImgEmb$ and $TextEmb$:

$$ImgEmb_{norm}(i,:) = \frac{ImgEmb(i,:)}{\|ImgEmb(i,:)\|} \tag{9}$$

$$TextEmb_{norm}(j,:) = \frac{TextEmb(j,:)}{\|TextEmb(j,:)\|} \tag{10}$$

We then compute the cosine similarity matrix:

$$S(i,j) = ImgEmb_{norm}(i,:) \cdot TextEmb_{norm}(j,:)^T \tag{11}$$

Where $S \in \mathbb{R}^{N_{img}\times N_{usr}}$, and $S(i,j)$ denotes the similarity between the i-th image token and the j-th text token. Then, we find the K image tokens with the highest similarity to the text tokens and record their indexes. Specifically, we first need to expand the similarity matrix S into a one-dimensional array S_{flat} with dimension $N_{img} \times N_{usr}$:

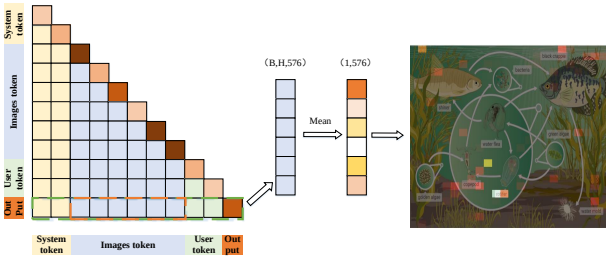


Figure 3: Influence rate of attention score about image tokens.

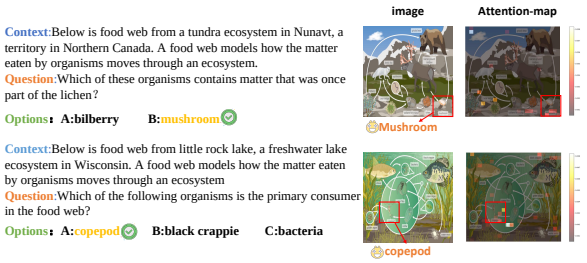


Figure 4: We find that the information flow converges in regions related to the option of prompt, such as mushroom and copepod.

$$S_{flat} = flatten(S) \tag{12}$$

Where the dimension of S_{flat} is $1 \times P$, $P = N_{img} \times N_{usr}$. The expanded one-dimensional array S_{flat} is then sorted and the K subscripts with the highest similarity are obtained:

$$Indices = argsort(S_{flat})[-K:] \tag{13}$$

Using these one-dimensional subscripts, we can determine the subscripts of the corresponding image token. Since each subscript i corresponds to a 2D coordinate $(\lfloor i/N_{usr} \rfloor, i\%N_{usr})$, we only need the $i//N_{usr}$ part to represent the subscript of the image token:

$$Indimg_i = \lfloor Indices_i / N_{usr} \rfloor \tag{14}$$

Then, we set the attention mask of the tokens that do not belong to $Indimg$ to 0:

$$\mathbf{M}_i^I = \begin{cases} 1 & \text{if } i \in Indimg \\ 0 & \text{else} \end{cases} \tag{15}$$

Finally, we splice the attention masks of the system token, image token, and user token:

$$\mathbf{M} = \mathbf{M}^S + \mathbf{M}^I + \mathbf{M}^U \tag{16}$$

Where $\mathbf{M}^S = ones \in \mathbb{R}^{1 \times N_{sys}}$, and $\mathbf{M}^U = ones \in \mathbb{R}^{1 \times N_{usr}}$.

4 Experiment

4.1 Dataset and implementation details

The ScienceQA (Lu et al., 2022) dataset is currently the only dataset available for complex reasoning and contains 21,208 Q&A multiple-choice questions from elementary and middle school science curricula. A typical question contains multimodal context, correct options, generalized background knowledge, and specific explanations. Our experiments were performed on a single 4090D GPU. We applied our approach to the 7b and 13b models of LLaVA-1.5, the Mipha-3B (Zhu et al., 2024) model. We used greedy search for all experiments. It is worth noting that we focus only on data containing images and only on Zero-shot Learning.

4.2 Attention convergence

When solving visual complex reasoning questions, humans tend to look for information in images that are related to the text. This behavior can efficiently obtain the key information of the image. To explore the degree of attention to image information during multimodal large model reasoning, as shown in Fig. 4, we obtained the attention scores of image tokens and superimposed them on the original image. Images will have a significant tendency to the options appearing in the cue, specifically, LLM will pay special attention to the word or object in the image that is closely related to the text information. This indicates that image interacts with text in the reasoning process, and MLLM is more inclined to pay attention to the image tokens that are related to the text.

4.3 Image-text similarity

We explored the relationship between image tokens and text tokens. We map embedding to the same cosine metric space, as shown in Fig. 5, and we find that there is a similarity between the image token and text token in the cosine metric space. Specifically, we find that the distribution of image tokens is similar to that of text tokens, and both of them are concentrated in the metric space of $X < 0$. In the blue rectangle, image tokens and text tokens are densely distributed and have strong similarities. Based on the attention convergence phenomenon

Table 1: Comparison among different MLLMs on ScienceQA benchmarks, "Res" represents the input image resolution.

Method	Learning	LLM backbone	Res	SQA(IMG)%
Instruct-BLIP (Dai et al., 2024)	Zero-shot	Vicuna-7B	224	60.50
Instruct-BLIP (Dai et al., 2024)	Zero-shot	Vicuna-13B	224	63.10
BLIP-2 (Li et al., 2023a)	Zero-shot	Vicuna-13B	224	61.03
Shikra (Chen et al., 2023)	Zero-shot	Vicuna-13B	224	45.80
DDCoT(GPT3.5) (Zheng et al., 2023)	Zero-shot	175B	-	72.53
DDCoT(MiniGPT-4) (Zheng et al., 2023)	Zero-shot	Vicuna-13B	-	56.72
Ying-VLM (Li et al., 2023b)	Zero-shot	-	-	55.70
Otter (Zhao et al., 2023)	Zero-shot	-	-	66.30
MiniGPT-4 (Zhu et al., 2023)	Zero-shot	Vicuna-13B	336	42.34
Qwen-VL-Chat (Bai et al., 2023)	Zero-shot	Qwen-7B	448	68.21
MobileVLM (Chu et al., 2023b)	Zero-shot	MobileLLaMA	336	61.00
Qwen-VL (Bai et al., 2023)	Zero-shot	Qwen-7B	448	67.12
Mipha-3B (Liu et al., 2023)	Zero-shot	Phi-2-2.7B	384	70.40
Mipha-3B+ours	Zero-shot	Phi-2-2.7B	384	**70.85**
LLaVA1.5 (Liu et al., 2023)	Zero-shot	Vicuna-7B	336	65.15
LLaVA1.5+ours	Zero-shot	Vicuna-7B	336	**68.02**
LLaVA1.5 (Liu et al., 2023)	Zero-shot	Vicuna-13B	336	72.09
LLaVA1.5+ours	Zero-shot	Vicuna-13B	336	**73.23**

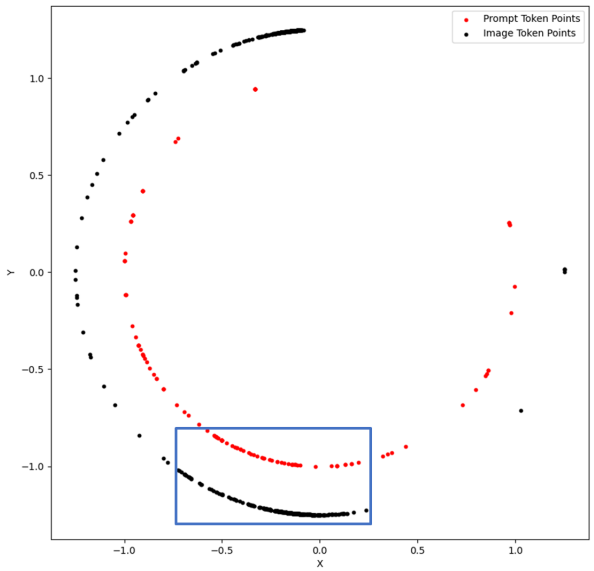


Figure 5: Distribution of image token and prompt token in cosine metric space.

in Section 4.2 and the similarity between image and text in the cosine metric space, we design a method for selecting image tokens with strong relevance to text so that MLLM obtains useful image tokens and avoids the influence of irrelevant tokens on MLLM reasoning.

We use different MLLMs for evaluation on the ScienceQA dataset, including the 7b and 13b models of LLaVA-v1.5, and the Mipha-3B model. As shown in Table 1, we show the results for the different baseline models as well as for the models to which our method is applied. As can be seen in Table 1, the models after applying our method have a great improvement compared to the baseline model, especially the LLaVA1.5-7B model improves the accuracy by 2.87%. In the experiments with ScienceQA (Image) as the dataset, our method is optimal in both Zero-shot Learning methods with Vicuna-7B or Vicuna-13B as the backbone.

4.4 Ablation study

In this section, we investigate the effects of ignoring different numbers of image tokens, ignoring image tokens with different levels of importance, and different similarity algorithms on the experimental results. In addition, we also present experiments ignoring text tags in Section A.0.1.

4.4.1 Ignore varying amounts of image tokens

To explore the effect of ignoring different numbers of image tokens on LLM's complex reasoning task, we set up ten sets of comparison experiments. As shown in Table 2, we obtained the accuracy as well as the running time of LLM on the ScienceQA (Image) dataset with different numbers of image tokens. Note that the LLM we use is LLaVA1.5-7B. We find that the time for LLM to perform a run is also reduced when some image tokens are ignored. The more image tokens are ignored, the less time is consumed. This indicates that when we ignore some unimportant image tokens, the

Table 2: Accuracy and runtime of LLM when ignoring different numbers of image tokens(baseline: LLaVA1.5-7B).

Ignored number	Accuracy(%)	Running time(/s)
72 /576	67.23	292
124 /576	**68.02**	279
144 /576	67.67	277
216 /576	67.28	265
288 /576	66.48	260
360 /576	66.04	255
432 /576	65.54	252
504 /576	63.96	246
576 /576	53.35	244
Baseline	65.15	303

original dense attention matrix becomes diluted, which greatly reduces the computation. As the number of ignored tokens increases, the attention matrix becomes more sparse and less computationally intensive. Interestingly, when we ignore all the image tokens, the accuracy of LLM still has 53.35%. However, since it is unlikely that all data have the same image, it is crucial to adaptively choose the appropriate number of image tokens to ignore, which will be our next research work.

4.4.2 The impact of image tokens of different importance on the inference ability

To investigate the importance of image tokens in improving the model's complex reasoning ability, we conducted an ablation study. As shown in Table 3, we conducted four sets of experiments, namely, ignoring unimportant image tokens (Experiment 1), ignoring image tokens of intermediate importance (Experiment 2), ignoring important image tokens (Experiment 3), and randomly ignoring image tokens (Experiment 4). In particular, the importance of an image token is calculated based on its similarity to text. In Experiment 2, we conducted ten sets of experiments and took the average as the final result. The results show that the best results were obtained in Experiment 1 and poor results were obtained in Experiment 3, indicating that important image tokens play a positive role in the complex reasoning task of LLM, while unimportant image tokens play a negative role in the complex reasoning task of LLM. The results of Experiment 2 are located between Experiment 1 and Experiment 3, indicating that the accuracy of LLM's complex reasoning task shows a positive correlation with the importance of image tokens. The results of Experiment 4 contain both higher and lower scores than Baseline. I believe that lower scores are obtained when some important image tokens are randomly ignored, and higher scores are obtained when some unimportant image tokens are randomly ignored. We provide detailed results of Experiment 4 in Section A.0.2 of the Supplementary Material.

Table 3: The effect of image tokens of varying importance on a complex reasoning task.

Ignored	Model	Ignored number	Accuracy(%)
Unimportant	LLaVA1.5-7B	124 /576	**68.02**
Intermediate	LLaVA1.5-7B	124 /576	64.55
Important	LLaVA1.5-7B	124 /576	61.73
Random	LLaVA1.5-7B	124 /576	65.12
Baseline	LLaVA1.5-7B	0 /576	65.15

4.4.3 Different similarity algorithms

To investigate the effect of different similarity algorithms on the experimental results, we conducted three different sets of experiments. We conducted experiments using cosine similarity, Euclidean distance similarity, and Manhattan distance similarity, and obtained the accuracy of the LLM when using different schemes. As shown in Table 4, we found the best improvement using the cosine similarity algorithm, and the other two algorithms also achieved a small improvement. This is because cosine similarity performs better in embedding similarity calculation tasks by measuring the angle between vectors instead of the absolute distance, which ignores scale differences and can better capture the semantic correlation between image and text embedding. In high-dimensional spaces, the distance between vectors tends to be uniform, and Euclidean and Manhattan distances cannot measure this similarity equally effectively because they are affected by the size and scale of the vectors.

Table 4: The effect of different algorithms for computing similarity on the accuracy of LLM complex reasoning.

Model	Algorithm	Number of ignored	Accuracy
LLaVA1.5-7B	Cosine Similarity	124	**68.02**
LLaVA1.5-7B	Euclidean Distance	124	66.73
LLaVA1.5-7B	Manhattan Distance	124	66.88
Baseline		0	65.15

4.5 Using cluster analysis on image token embedding

To further explore how our work improves LLM answer accuracy, we performed some analysis on image data. Take the question with id 1879 in the ScienceQA dataset as an example. Fig. 6 shows

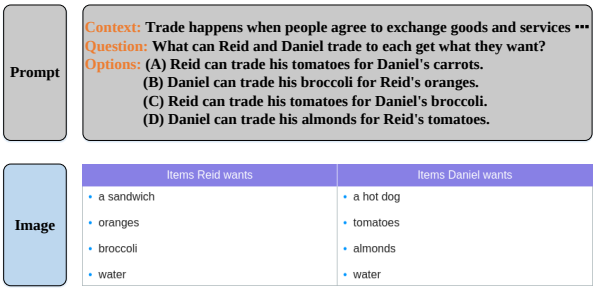
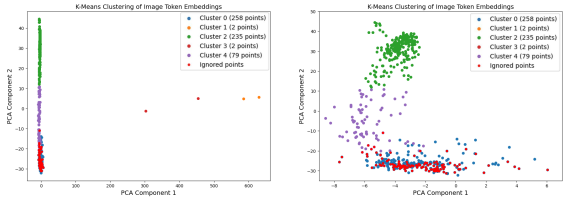


Figure 6: This is a question closely related to the information in the picture and the correct answer to this question is 'C'.

the prompt of the question with the image. In the baseline method, LLM gave the wrong answer. After using our proposed method, LLM successfully answered the correct answer. To investigate why LLM can correctly infer the answer after ignoring some image tokens, we use k-means clustering for image embedding, where the embedding tensor of each image token is mapped as a point in the 2D plane. Fig. 7 shows the embedding of the 576 image markers mapped to points on the 2D plane after clustering. The red points indicate the mapped points of the image tokens that we ignore. We notice that the image tokens ignored by our method are concentrated in cluster0, so we ignore all 258 image tokens in cluster0 and still get the correct answer. In addition, we also did a comparison experiment, as shown in Table 5. By analyzing the above experiments, we propose some conjectures: there are some 'spy' tokens in cluster0, which affect LLM's understanding of the image and cause answering errors. Cluster2 and cluster4 contain some important tokens, which will also affect LLM's understanding of the image if ignored.



(a) (b)

Figure 7: Fig.(a) represents the distribution of image tokens. Where the red dots indicate the distribution position of tokens which are ignored according to the similarity, Fig.(b) is a localized zoomed-in view of Fig.(a).

To verify our conjecture, we found a critical value: 86. We find that ignoring any number less than 86 causes MLLM to answer incorrectly while ignoring a number not less than 86 causes MLLM to answer correctly, and these 86 tokens are all located in cluster0, which suggests that the 86 tokens located in cluster0 will not improve MLLM's comprehension of the image, but rather play a negative role in the accuracy of MLLM's answer. We analyzed more data and came to similar conclusions. In addition, we found two common phenomena, for which we give details in Section A.0.3.

Table 5: The answer to the LLM response is obtained by ignoring the image tokens in different clusters. ✔ means ignored, ✗ means do not ignored.

cluster0	cluster1	cluster2	cluster3	cluster4	Result
✗	✗	✗	✗	✗	F
✔	✔	✗	✗	✗	T
✔	✗	✔	✗	✗	T
✔	✗	✗	✔	✗	T
✔	✗	✗	✗	✔	T
✗	✔	✗	✗	✗	F
✗	✗	✔	✗	✗	F
✗	✗	✗	✔	✗	F
✗	✗	✗	✗	✔	F
✗	✗	✔	✗	✔	T

5 Conclusion and discussion

This study proposes an innovative method to enhance the performance of MLLM in complex reasoning tasks by computing the similarity between image and text embeddings. We find that in the LLM decoder, image tokens semantically related to text are more likely to converge information flows. Based on this finding, we designed the Simignore algorithm, which improves the similarity between computed images and text embeddings and filters out irrelevant image information. Through extensive experiments, we demonstrate that Simignore improves complex reasoning for MLLMs of different sizes.

Although Simignore has achieved significant results in improving the inference capabilities of MLLM, there are still some limitations. Future work will explore strategies for adaptively choosing the number of image tokens to ignore, as well as investigate the internal mechanisms of the model to more fully understand how image and textual information work together to facilitate complex reasoning.

References

Jinze Bai, Shuai Bai, Shusheng Yang, Shijie Wang, Sinan Tan, Peng Wang, Junyang Lin, Chang Zhou, and Jingren Zhou. 2023. Qwen-vl: A versatile vision-language model for understanding, localization, text reading, and beyond.

Tom Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared D Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, et al. 2020. Language models are few-shot learners. *Advances in neural information processing systems*, 33:1877–1901.

Keqin Chen, Zhao Zhang, Weili Zeng, Richong Zhang, Feng Zhu, and Rui Zhao. 2023. Shikra: Unleashing multimodal llm's referential dialogue magic. *arXiv preprint arXiv:2306.15195*.

Liang Chen, Haozhe Zhao, Tianyu Liu, Shuai Bai, Junyang Lin, Chang Zhou, and Baobao Chang. 2024. An image is worth 1/2 tokens after layer 2: Plug-and-play inference acceleration for large vision-language models. *arXiv preprint arXiv:2403.06764*.

Xiangxiang Chu, Limeng Qiao, Xinyang Lin, Shuang Xu, Yang Yang, Yiming Hu, Fei Wei, Xinyu Zhang, Bo Zhang, Xiaolin Wei, et al. 2023a. Mobilevlm: A fast, reproducible and strong vision language assistant for mobile devices. *arXiv preprint arXiv:2312.16886*.

Xiangxiang Chu, Limeng Qiao, Xinyang Lin, Shuang Xu, Yang Yang, Yiming Hu, Fei Wei, Xinyu Zhang, Bo Zhang, Xiaolin Wei, et al. 2023b. Mobilevlm: A fast, reproducible and strong vision language assistant for mobile devices. *arXiv preprint arXiv:2312.16886*.

Xiangxiang Chu, Limeng Qiao, Xinyu Zhang, Shuang Xu, Fei Wei, Yang Yang, Xiaofei Sun, Yiming Hu, Xinyang Lin, Bo Zhang, et al. 2024. Mobilevlm v2: Faster and stronger baseline for vision language model. *arXiv preprint arXiv:2402.03766*.

Wenliang Dai, Junnan Li, Dongxu Li, Anthony Meng Huat Tiong, Junqi Zhao, Weisheng Wang, Boyang Li, Pascale N Fung, and Steven Hoi. 2024. Instructblip: Towards general-purpose vision-language models with instruction tuning. *Advances in Neural Information Processing Systems*, 36.

Haiwen Diao, Ying Zhang, Lin Ma, and Huchuan Lu. 2021. Similarity reasoning and filtration for image-text matching. In *Proceedings of the AAAI conference on artificial intelligence*, volume 35, pages 1218–1226.

Fartash Faghri, David J Fleet, Jamie Ryan Kiros, and Sanja Fidler. 2017. Vse++: Improving visual-semantic embeddings with hard negatives. *arXiv preprint arXiv:1707.05612*.

Andrea Frome, Greg S Corrado, Jon Shlens, Samy Bengio, Jeff Dean, Marc'Aurelio Ranzato, and Tomas Mikolov. 2013. Devise: A deep visual-semantic embedding model. *Advances in neural information processing systems*, 26.

Ryan Kiros, Ruslan Salakhutdinov, and Richard S Zemel. 2014. Unifying visual-semantic embeddings with multimodal neural language models. *arXiv preprint arXiv:1411.2539*.

Takeshi Kojima, Shixiang Shane Gu, Machel Reid, Yutaka Matsuo, and Yusuke Iwasawa. 2022. Large language models are zero-shot reasoners. *Advances in neural information processing systems*, 35:22199–22213.

Junnan Li, Dongxu Li, Silvio Savarese, and Steven Hoi. 2023a. Blip-2: Bootstrapping language-image pre-training with frozen image encoders and large language models. In *International conference on machine learning*, pages 19730–19742. PMLR.

Junnan Li, Dongxu Li, Caiming Xiong, and Steven Hoi. 2022. Blip: Bootstrapping language-image pre-training for unified vision-language understanding and generation. In *International conference on machine learning*, pages 12888–12900. PMLR.

Lei Li, Yuwei Yin, Shicheng Li, Liang Chen, Peiyi Wang, Shuhuai Ren, Mukai Li, Yazheng Yang, Jingjing Xu, Xu Sun, et al. 2023b. M^{3}it: A large-scale dataset towards multi-modal multilingual instruction tuning. *arXiv preprint arXiv:2306.04387*.

Haotian Liu, Chunyuan Li, Yuheng Li, and Yong Jae Lee. 2023. Improved baselines with visual instruction tuning. *arXiv preprint arXiv:2310.03744*.

Yang Liu, Lei Zhou, Xiao Bai, Yifei Huang, Lin Gu, Jun Zhou, and Tatsuya Harada. 2021. Goal-oriented gaze estimation for zero-shot learning. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 3794–3803.

Pan Lu, Swaroop Mishra, Tanglin Xia, Liang Qiu, Kai-Wei Chang, Song-Chun Zhu, Oyvind Tafjord, Peter Clark, and Ashwin Kalyan. 2022. Learn to explain: Multimodal reasoning via thought chains for science question answering. *Advances in Neural Information Processing Systems*, 35:2507–2521.

Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. 2021. Learning transferable visual models from natural language supervision. In *International conference on machine learning*, pages 8748–8763. PMLR.

Alec Radford, Jeffrey Wu, Rewon Child, David Luan, Dario Amodei, Ilya Sutskever, et al. 2019. Language models are unsupervised multitask learners. *OpenAI blog*, 1(8):9.

Yuzhang Shang, Mu Cai, Bingxin Xu, Yong Jae Lee, and Yan Yan. 2024. Llava-prumerge: Adaptive token reduction for efficient large multimodal models. *arXiv preprint arXiv:2403.15388*.

Weihan Wang, Qingsong Lv, Wenmeng Yu, Wenyi Hong, Ji Qi, Yan Wang, Junhui Ji, Zhuoyi Yang, Lei Zhao, Xixuan Song, et al. 2023. Cogvlm: Visual expert for pretrained language models. *arXiv preprint arXiv:2311.03079*.

Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Fei Xia, Ed Chi, Quoc V Le, Denny Zhou, et al. 2022. Chain-of-thought prompting elicits reasoning in large language models. *Advances in neural information processing systems*, 35:24824–24837.

Jianwei Yang, Chunyuan Li, Pengchuan Zhang, Bin Xiao, Ce Liu, Lu Yuan, and Jianfeng Gao. 2022. Unified contrastive learning in image-text-label space. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 19163–19173.

Zhengqing Yuan, Zhaoxu Li, and Lichao Sun. 2023. Tinygpt-v: Efficient multimodal large language model via small backbones. *arXiv preprint arXiv:2312.16862*.

Ruigeng Zeng, Wentao Ma, Xiaoqian Wu, Wei Liu, and Jie Liu. 2024. Image–text cross-modal retrieval with instance contrastive embedding. *Electronics*, 13(2):300.

Xiaohua Zhai, Basil Mustafa, Alexander Kolesnikov, and Lucas Beyer. 2023. Sigmoid loss for language image pre-training. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 11975–11986.

Haozhe Zhao, Zefan Cai, Shuzheng Si, Xiaojian Ma, Kaikai An, Liang Chen, Zixuan Liu, Sheng Wang, Wenjuan Han, and Baobao Chang. 2023. Mmicl: Empowering vision-language model with multi-modal in-context learning. *arXiv preprint arXiv:2309.07915*.

Ge Zheng, Bin Yang, Jiajin Tang, Hong-Yu Zhou, and Sibei Yang. 2023. Ddcot: Duty-distinct chain-of-thought prompting for multimodal reasoning in language models. *Advances in Neural Information Processing Systems*, 36:5168–5191.

Deyao Zhu, Jun Chen, Xiaoqian Shen, Xiang Li, and Mohamed Elhoseiny. 2023. Minigpt-4: Enhancing vision-language understanding with advanced large language models. *arXiv preprint arXiv:2304.10592*.

Minjie Zhu, Yichen Zhu, Xin Liu, Ning Liu, Zhiyuan Xu, Chaomin Shen, Yaxin Peng, Zhicai Ou, Feifei Feng, and Jian Tang. 2024. A comprehensive overhaul of multimodal assistant with small language models. *arXiv preprint arXiv:2403.06199*.

A Example Appendix

A.0.1 Ignore text token instead of image token

To explore the effect of text tokens on model accuracy, we refer to the method of ignoring image tokens and experiment with ignoring text tokens. The LLM we use is LLaVA1.5-7B. The experimental results are shown in Table 6, where the randomized trials are the average of ten experimental results. The specific experimental results are shown in Table 7. We were surprised to find that appropriately ignoring some unimportant text tokens can also improve the accuracy of LLM in complex reasoning tasks. However, text seems to be more important than image, because the accuracy obtained by ignoring only 25% of text tokens is very close to the accuracy obtained by ignoring 100% of image tokens. In ten experiments of randomly selecting text tokens, the results of all the experiments are lower than the results of Baseline, which indicates that the conditions for selecting the correct text token for ignoring are very demanding, and further illustrates the importance of text tokens compared to image tokens.

Table 6: Impact of ignoring text tokens on LLM complex reasoning. SFH2L means Select from highest to lowest importance, RS means Randomly select. RS is the average of ten experiments.

Model	Strategy	Percentage ignored	Accuracy(%)
LLaVA1.5-7B	SFH2L	5.0	66.58
LLaVA1.5-7B	SFH2L	12.5	63.81
LLaVA1.5-7B	SFH2L	25.0	52.35
LLaVA1.5-7B	RS	5.0	59.59
Baseline		0	65.15

A.0.2 Results of ten sets of experiments in which 124 image tokens were randomly ignored

Table 8 shows the experimental results of the ten groups of randomized trials in Section 4.4.2. We find that out of these ten sets of experiments, five sets of experiments have results above the baseline and the remaining five sets have results below the baseline. We believe that when the 124 image tokens randomly ignored contain a higher number of text-independent tokens, then it will help the LLM to make the right choices. Including a large number of tokens that are not relevant to the text will prevent LLM from making correct judgments.

Table 7: Results of ten sets of experiments in which 5% text markup was randomly ignored

Experiment No.	Accuracy(%)
1	61.18
2	59.44
3	59.00
4	59.30
5	59.20
6	60.24
7	59.35
8	59.00
9	59.25
10	59.89
Average	59.59
Baseline	65.15

Table 8: Results of ten sets of experiments in which 124 image tokens were randomly ignored

Experiment No.	Accuracy(%)
1	64.70
2	65.00
3	64.60
4	65.29
5	65.34
6	65.64
7	65.54
8	64.95
9	64.70
10	65.44
Average	65.12
Baseline	65.15

A.0.3 Two common phenomena that occur after clustering of image tokens

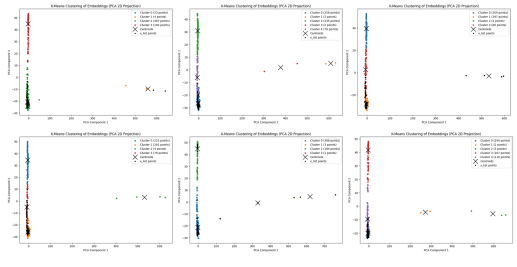


Figure 8: A few points deviate from the rest.

In Section 4.5, we downscale image tokens to points in the 2D plane and then cluster these points. We analyzed more data in the ScienceQA(Image) dataset and found two common phenomena:

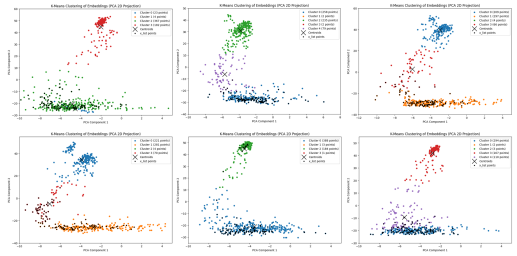


Figure 9: The ignored tokens are almost concentrated in the cluster with the smallest Y value.

1. As shown in Fig. 8, we found that there are always about 4 points in the 2D plane after clustering that are separated from other points. 2. As shown in Fig. 9, we found that the points we ignored for the image token mapping to the 2D plane (the black points) are almost distributed in the cluster with the lowest Y-value. When we ignore the token corresponding to these points, we can improve the accuracy of the LLM answer, and this finding further validates our conjecture about the 'spy' token.